

Sri Sai Rama Krishna Rao Mandapaka

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PROFESSIONAL SUMMARY

Civil Engineering Master's graduate with field engineering experience performing construction inspection, materials testing, and QA/QC documentation on a \$130 million rural water infrastructure program (300 villages, Jal Jeevan Mission) and a 973 MW underground power generation facility (178 meters below surface, Gayatri Pump House). Experienced verifying compliance of rebar placement, concrete and asphalt work, drainage structures, and subgrade/base construction against project plans and specifications; performing soil density and concrete material testing; and preparing diary inspection reports, QA/QC checklists, and as-built documentation. Procore Certified: Project Manager (Core Tools). OSHA 30-Hour, CMIT, CPR and AED certified. FE Exam scheduled for October 2026. Authorized to work in the U.S. Open to relocation nationwide.

TECHNICAL SKILLS

Field Execution: Site Inspection, QA/QC Inspections, Soil/Concrete/Asphalt Material Testing and Inspection, Field Surveying and Layout, Reinforcing Steel and Concrete Form Inspection, Production Rate Tracking, Construction Safety, Diary/Daily Inspection Reports, Nonconformance Documentation

Project Engineering and Controls: RFI Management, Submittal Coordination, Labor Control, Pay Application Verification, Change Order Tracking, WIP Reporting, CPM Scheduling, Lean Construction Practices, Look-Ahead Scheduling, Punch List Management, Closeout Documentation, Subcontractor Coordination, Budget Reconciliation, Lump Sum and Unit-Price Contracts

BIM and Digital Construction: Revit Modeling, Navisworks Clash Detection, BIM Coordination, 4D Scheduling, MEP Coordination, Model-Based Constructability Analysis

Software: Procore, Primavera P6, MS Project, Bluebeam, AutoCAD, Revit, Navisworks, Microsoft Office Suite (Excel, Word, PowerPoint, Outlook), On-Screen Takeoff (OST), Civil 3D (academic)

Codes and Standards: AISC 360, ACI 318, ASCE 7, IBC, IS 456, IS 800

CERTIFICATIONS

Procore Certified: Project Manager (Core Tools) | Lean Construction, Procore Academy | CPR and AED Essentials | Permit-Required Confined Spaces for Construction | OSHA 30-Hour Construction Safety | CMIT (Construction Manager in Training), CMCI (Valid through April 2033) | FE Exam (Fundamentals of Engineering), Scheduled October 2026

PROFESSIONAL EXPERIENCE

Site Engineer

Dec 2022 – Dec 2023

Viswanath Projects Ltd, Jal Jeevan Mission | Bahraich, Uttar Pradesh, India

- Observed daily construction activities across multiple concurrent sites and verified work was performed in substantial compliance with contract plans and specifications, reporting deficiencies to the Project Superintendent.
- Verified installed quantities of pipeline, RCC, and foundation works against billing statements using Microsoft Excel; ensured pay-for-performance accuracy on contracts exceeding \$1.5 million and prevented overbilling.
- Managed punch lists and prepared closeout documentation including as-built records, QA/QC checklists, measurement books, and audit-compliant project files; confirmed completed work complied with applicable codes, drawings, and specifications.
- Performed labor control on assigned project phases, including daily tracking of labor productivity and unit man-hour performance; compiled weekly labor reports for Project Manager review and budget reconciliation.
- Managed RFI and submittal logs between field teams, subcontractors, and project management; maintained complete communication records and ensured timely resolution to support CPM schedule compliance.
- Administered change order tracking and prepared cost variance reports; coordinated with the Project Manager to deliver up-to-date financial information for budget control.
- Developed and maintained CPM project schedules using Primavera P6; prepared short-term look-ahead schedules and applied Lean construction techniques including weekly work plans and constraint elimination.

Key Skills: Construction QA/QC Inspection, Pay Application Verification, Labor Control & Productivity Tracking,

RFI/Submittal Management, Change Order Tracking, Primavera P6 Scheduling, Lean Construction, Closeout Documentation

Graduate Site Intern Supervisor

Jan 2022 – Nov 2022

Megha Engineering and Infrastructures Ltd, Gayatri Pump House, Lift Irrigation Project | Karimnagar, Telangana, India

- Conducted QA/QC inspections and material testing per project specifications; inspected reinforcing steel placement and embedded items prior to concrete placement, maintained quality assurance logs, and reported nonconformances for resolution.
- Coordinated daily field activities across active underground construction zones at a 973 MW power generation facility 178 meters below surface, reporting directly to the Project Superintendent.
- Reviewed and verified contractor invoices and subcontractor bills against measured quantities and contract rates; ensured accurate pay application processing and prevented overbilling on this mega civil infrastructure project.
- Performed labor control and production tracking across assigned underground construction phases; tracked unit man-hour performance and maintained production rate reports for the project team.
- Managed RFI and submittal coordination; maintained drawing logs with revision numbers and issuance dates, resolved document conflicts, and ensured the most current drawings were distributed to all field teams.

Key Skills: QA/QC Inspection & Material Testing, Reinforcing Steel Inspection, Underground Construction Coordination, Pay Application Review, Labor & Production Tracking, RFI/Submittal Coordination, Drawing Log Management

KEY ACADEMIC PROJECTS

Non-Thesis Research: Impact of Core BIM Elements on Cost and Productivity (University of Alabama at Birmingham, 2025)

- Built a federated BIM model using Revit and Navisworks, resolving over 300 coordination clashes across architectural, structural, and MEP systems; quantified 18–27% cost savings, 20–25% productivity gains, and 30–40% material wastage reduction through comparative analysis.

Construction Estimation and Bidding: Residential Garage Project (University of Alabama at Birmingham, 2024)

- Developed a complete Bill of Quantities and cost estimate using On-Screen Takeoff and RS Means unit pricing; performed plan and specification review and built a comparative bid analysis model to support subcontractor solicitation and award.

EDUCATION

Master of Science, Civil Engineering (ABET Accredited) — GPA: 3.60 / 4.0

Jan 2024 – Dec 2025

University of Alabama at Birmingham | Birmingham, AL

Bachelor of Technology, Civil Engineering — CGPA: 8.7 / 10

Jul 2018 – Apr 2022

RGUKT IIIT Srikakulam, Rajiv Gandhi University of Knowledge Technologies | India